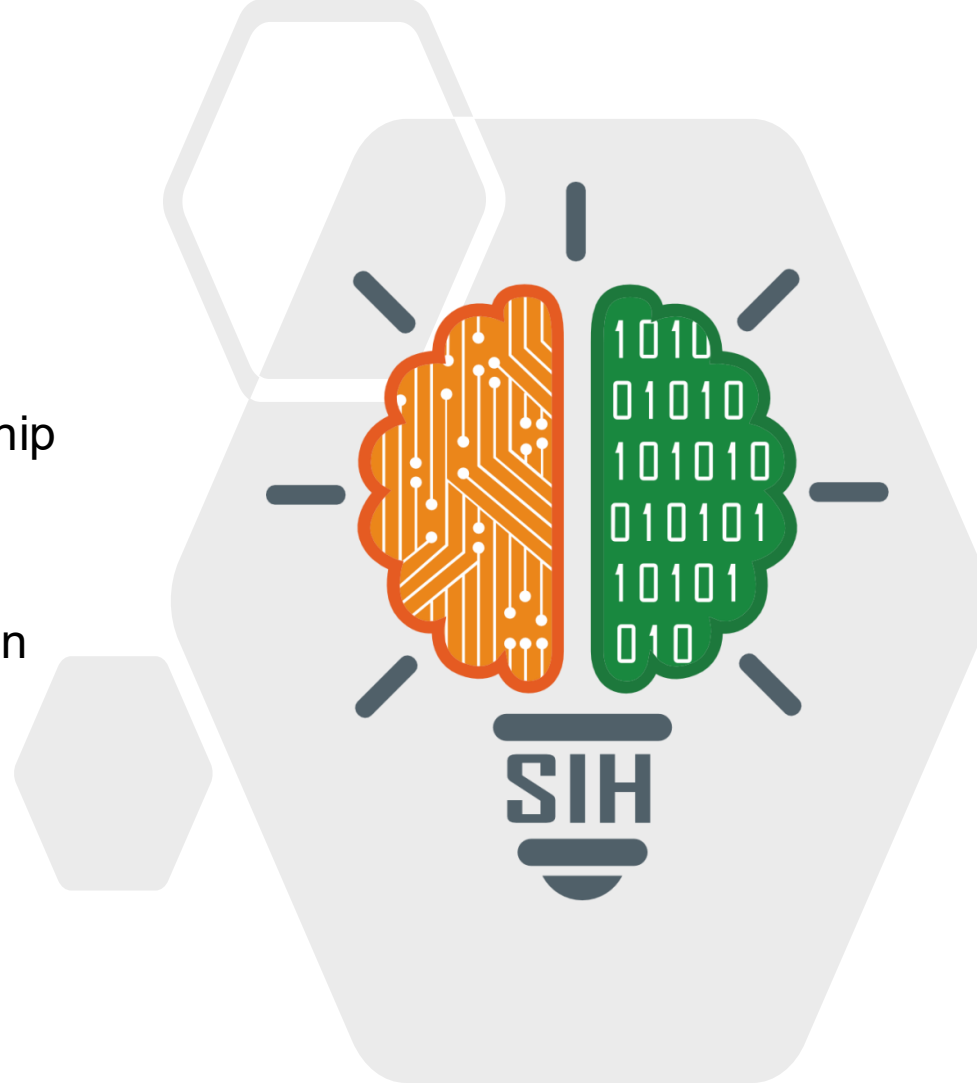


SMART INDIA HACKATHON 2025



- **Problem Statement ID** –
- **Problem Statement Title** – Secure Authorization & Fraud Prevention for Land Ownership Transactions
- **Theme** – GovTech / Cybersecurity / Blockchain
- **PS Category** – Software
- **Team ID** –



Detailed Explanation of Solution

- Every property is issued a Unique Land Parcel ID.
- Every ownership-changing transaction passes through: Owner Authentication → Document Verification → Fraud/Conflict Check → Government Authorization → Multi-Signature → Blockchain Commit.
- Two-key authorization: OWNER KEY + GOVERNMENT KEY = valid transaction. Neither party alone can execute a transfer.
- Sale Deed → SHA-256 hash → recorded on-chain; the actual document stays off-chain.

How it addresses the problem?

- Blocks forged documents, double sale, and unauthorized mutation via multi-party checks before commit.
- Fraud engine verifies seller = recorded owner, valid hash, valid signature, no mortgage/dispute hold, no duplicate Land Parcel ID.
- Illustrative scenario: Rahul → Fraudster attempt fails owner + government authorization → TRANSACTION BLOCKED.

Innovation and Uniqueness

BHUMI-VAULT does not reinvent blockchain land records. It adds the security and authorization layer that decides whether a transaction is trustworthy.

Land Parcel ID → Owner Auth → Document Verify → Fraud Check → Gov. Authorization → Multi-Sig → Blockchain → Ownership History Updated

Technologies to be used

Frontend

React / Next.js

Backend

Node.js + Express, REST APIs

Database

PostgreSQL — users, property metadata, transaction states, access logs

Documents

Off-chain secure storage for actual PDFs

Blockchain

Hyperledger Fabric — permissioned network

Cryptography

SHA-256 document hashing, digital signatures

Methodology & Process (Implementation Flow)

Citizen / Government User — React / Next.js UI

Node.js + Express Backend — REST APIs

PostgreSQL + Off-chain Storage — Identity & Key Service

SHA-256 Document Hash — Document integrity

Policy Engine / Chaincode — Rule gates

Hyperledger Fabric — Permissioned ledger

Department Nodes — Revenue | Registration | Survey | Bank | Judiciary

BHUMI-VAULT does not require replacing existing land-record infrastructure. It integrates as a security, verification and authorization layer on top of DILRMP, NGDRS, Land Records, Registration, Cadastral/Survey and Court records.

TECHNOLOGY

Mature, widely adopted open technologies: React, Node.js, PostgreSQL, Hyperledger Fabric.

INTEGRATION

API-based connection to existing government systems — no rip-and-replace needed.

PROTOTYPE

District-level simulated environment for demonstration and testing.

SCALABILITY

Permissioned architecture suited to multi-department participation across India.

Real-world data-quality caveat: Legacy Records → Government Verification → Data Cleansing → Digital Signing → Genesis Blockchain Record.
“Blockchain protects trusted history; it does not magically validate bad initial data.”

Potential challenges & risks

- Legacy / unreliable records
- Data privacy
- Large document sizes
- Lost private key
- Disputed property
- Insider manipulation
- Interoperability
- Rural connectivity

Mitigation strategy

- Verified genesis records after cleansing
- Permissioned blockchain + role-based access
- Off-chain storage + on-chain hashes only
- Government-assisted recovery process
- Automatic property freeze
- Multi-party signatures + immutable audit trail
- Standard APIs + shared schema
- Lightweight interface + assisted service centers

IMPACTS AND BENEFITS

One trusted property history. Multiple stakeholders.



Each event is timestamped · signed · hash-linked · auditable

LANDOWNER

- Unauthorized-transfer protection
- Permanent, tamper-evident history
- Easier ownership verification

GOVERNMENT

- Tamper-evident audit trail
- Faster mutation workflows
- Easier suspicious-transaction review

BANK

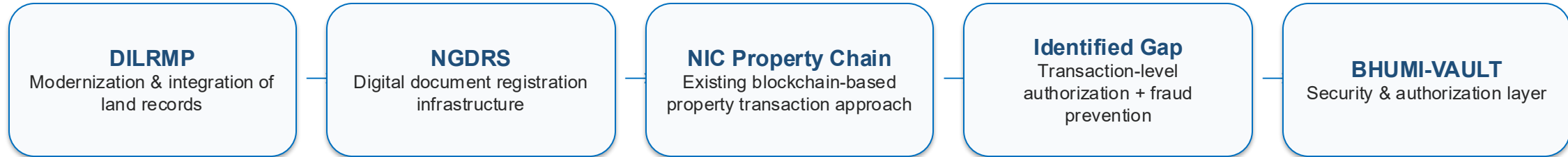
- Verify ownership instantly
- Check mortgage / encumbrance
- Check dispute status before lending

JUDICIARY

- Retrieve linked property history
- Trace ownership-changing events
- Less fragmented document investigation

Cryptographic authorization → Fewer unauthorized transactions → Higher trust → Faster verification → Better access to property-linked services

Research → Gap → BHUMI-VAULT



Details of the reference and research work

1. Department of Land Resources — DILRMP

National program modernizing and integrating land record digitization across states.

2. National Generic Document Registration System (NGDRS)

Standardized digital registration infrastructure for property documents.

3. NIC Centre of Excellence — Property Chain

Existing government blockchain approach for property transactions; BHUMI-VAULT builds the authorization layer on top of it.

4. Property Chain — Concept Paper

Defines the base blockchain land-record model referenced in this proposal.

5. NIC Land Records — Case Study

Documents real deployment learnings that shaped BHUMI-VAULT's integration approach.

6. Supreme Court of India — 2025 Judgment on Registration & Title

Legal grounding for how registration and title verification are treated in Indian law.